

An-Sheng (Anson) LU

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Electrical Engineering | Embedded Systems · Robotics · Controls | Seeking Summer 2027 Engineering Internship

EDUCATION

University of California, Riverside, B.S. Electrical Engineering **Expected Spring 2029**

Cumulative GPA: 3.87 | UCR Achievement Scholarship (2025–29), Dean's Honors List

Relevant Coursework: Data Structures & Algorithms (C++), Logic Design, Physics, Multivariable Calculus, ODE.

Victoria Academy, Yunlin, Taiwan, IB Bilingual Diploma **2022 – 2025**

GPA: 3.94 (IBDP) / 3.86 (Cumulative) | HL subjects: Mathematics AA (7), Physics (6), Chemistry (6).

Awards: Chairman's/Mayor's Award, Dr. Cho Rui-Chiao Scholarship, Excellence Award in Mathematics, School Scholarships (5x recipient, 2022–2024).

EXPERIENCES

Research Assistant, Robotics and Medical Systems (RaMS) Lab, UC Riverside **Oct. 2025 – Present**

- Awarded **UCR Research Mini-Grant** for an independently developed **computer vision/deep learning pipeline** that automates biomass estimation in collaboration with UCR Entomology Dept.; achieved **~90% biomass-estimation accuracy across 40 batches**.
- Built automated image-analysis workflows using Python — normal-distribution filtering, post-processing — that convert raw image data into phenotypic measurements for life-cycle research; implementation available on this [GitHub Repository](#).

Intern, Electrical Subsystems, Highlander Racing (FSAE), UC Riverside **Apr. 2026 – Present**

- Designing **industrial automation PCBs** in **Altium Designer** — from schematic capture of analog sensing stages to routing power stages.

Research Assistant, Biomechanics and Biomaterials Design Laboratory, UC Riverside **May 2026 – Aug. 2026**

- Diagnosed and resolved flow instability in a **Right Heart Simulator** (Physical Flow Testbed) by restructuring circuit schematics and running hydrodynamic diagnostics
- Designed the wire layout for an Arduino Uno R3 shield and custom **SolidWorks enclosures** — including a redesigned atrial reservoir chamber — improving structural integrity and fluid containment.

Intern, Robotics and Mechatronics Center, German Aerospace Center (DLR), Oberpfaffenhofen, Germany **Aug. 1 – 30, 2024**

- Designed a custom robotic system using an **OpenManipulator-X** robotic arm — system design, 3D-printed structural interfaces, motion control.
- Modeled mass-spring-damper systems in **MATLAB/Simulink**, designing a critically damped feedback controller to optimize system stability.
- Designed and assembled a **3x3x3 LED matrix** using an Arduino Micro; implemented a 4-mode firmware system featuring real-time pace control via ADC and a variable resistor.

Robotics Club Member, National Yunlin Univ. of Sci. & Tech. (NYUST), Yunlin, Taiwan **Sep. 2023 – June 2024**

- Collaborated with club members to program an omnidirectional mobile robot with a **three-Mecanum-wheel chassis**; executed sensor fusion using Ultrasonic and Color Detection sensors for obstacle avoidance and environmental interaction via LabVIEW.
- Implemented basic ROS interfacing for **system communication**.

PATENTS & INVENTIONS

Baton Device with Directional Indication, Taiwan Invention Patent No. I894025, Taiwan IP Office **June 2024 – Aug. 2025**

- Independently designed a signaling device with custom hardware logic using an **ESP32** and **MPU-6050 (IMU)** to process real-time indication; managed conceptual design and **circuit prototyping** through **patent approval**. [Taiwan Patent Search System](#).

HONORS & AWARDS

- 1st Place Winner, Robotics Competition, Yunlin, Taiwan, Dec. 25, 2023.
- 1st Place Winner, Industrial Robot Competition, Kaohsiung, Taiwan, Mar. 9, 2024.
- Final Shortlisted Prize, Electrical Engineering Category, 2024 All Ring Competition, Tainan, Taiwan, Oct. 24, 2024

SKILLS

- Language:** Mandarin (Native), English (Fluent).
- Programming:** C/C++, Python (PyTorch, OpenCV), LabVIEW, MATLAB/Simulink, ROS (system communication)
- Software & Hardware Skills:** Tinkercad, Creo, Arduino (IDE), Visual Studio/VS Code, Altium Designer, SolidWorks.